

RELATIONS & FUNCTIONS

NCERT + BOARD EXAM SHORT REVISION NOTES

Learn the Concept → Remember the Formula → Solve the Question

1. Cartesian Product

CONCEPT

If A and B are two non-empty sets, then the Cartesian product $A \times B$ is the set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

KEY RESULT / FORMULA

If

$$n(A) = m, \quad n(B) = n,$$

then

$$n(A \times B) = mn$$

In general,

$$A \times B \neq B \times A.$$

BOARD ALERT

$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

Also,

$$A \times B = \emptyset$$

if either $A = \emptyset$ or $B = \emptyset$.

2. Relation

CONCEPT

A relation R from a non-empty set A to a non-empty set B is a subset of $A \times B$.
Thus,

$$R \subseteq A \times B$$

If $(a, b) \in R$, we write

$$aRb.$$

KEY RESULT / FORMULA

For a relation R from A to B :

$$\text{Domain of } R = \{a \in A : \text{there exists } b \in B \text{ such that } (a, b) \in R\}$$

$$\text{Range of } R = \{b \in B : \text{there exists } a \in A \text{ such that } (a, b) \in R\}$$

The remaining elements of B form the **co-domain minus range**.

3. Number of Relations

Suppose

$$n(A) = m, \quad n(B) = n.$$

Then

$$n(A \times B) = mn.$$

Since every subset of $A \times B$ can be a relation,

KEY RESULT / FORMULA

$$\text{Number of relations from } A \text{ to } B = 2^{mn}$$

4. Types of Relations

Let R be a relation on a set A .

Reflexive Relation

$$(a, a) \in R \quad \forall a \in A$$

Every element must be related to itself.

QUICK EXAMPLE

If

$$A = \{1, 2, 3\},$$

then a reflexive relation must contain

$$(1, 1), (2, 2), (3, 3).$$

Symmetric Relation

$$(a, b) \in R \Rightarrow (b, a) \in R$$

Transitive Relation

$$(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$$

Equivalence Relation

A relation is an equivalence relation if it is:

Reflexive + Symmetric + Transitive

BOARD ALERT

To prove an equivalence relation, check all three properties separately.

5. Function

CONCEPT

A relation f from A to B is called a function if:

Every element of A has exactly one image in B .

We write

$$f : A \rightarrow B.$$

If

$$f(a) = b,$$

then b is called the image of a .

BOARD ALERT

For a function:

Each input has exactly ONE output.

But different inputs may have the same output.

QUICK EXAMPLE

$$f = \{(1, 2), (2, 4), (3, 4)\}$$

is a function because each first component occurs exactly once.

But

$$g = \{(1, 2), (1, 3), (2, 4)\}$$

is **not** a function because 1 has two images.

6. Domain, Codomain and Range

For

$$f : A \rightarrow B,$$

$$\text{Domain} = A$$

$$\text{Codomain} = B$$

$$\text{Range} = \{f(x) : x \in A\}$$

Always remember:

$$\text{Range} \subseteq \text{Codomain}$$

7. Types of Functions

One-One Function (Injective)

A function $f : A \rightarrow B$ is one-one if

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

Equivalent condition:

$$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2).$$

Many-One Function

If two or more distinct elements of A have the same image in B , then f is many-one.

$$x_1 \neq x_2, \quad f(x_1) = f(x_2).$$

Onto Function (Surjective)

A function $f : A \rightarrow B$ is onto if every element of B has at least one pre-image in A .

$$\text{Range} = \text{Codomain}$$

Into Function

A function is into if at least one element of the codomain is not an image of any element of the domain.

$$\text{Range} \subsetneq \text{Codomain}$$

Bijjective Function

A function is bijective if it is both one-one and onto.

$$\text{Bijective} = \text{One-One} + \text{Onto}$$

8. Important Special Functions

Identity Function

For $f : A \rightarrow A$,

$$f(x) = x$$

for every $x \in A$.

It is both one-one and onto.

Constant Function

$$f(x) = c$$

for all x in the domain.

If the domain has more than one element, a constant function is many-one.

9. Algebra of Functions

If

$$f : A \rightarrow \mathbb{R}, \quad g : A \rightarrow \mathbb{R},$$

then

$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x)$$

and, where $g(x) \neq 0$,

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

10. Composition of Functions

If

$$f : A \rightarrow B, \quad g : B \rightarrow C,$$

then the composite function

$$g \circ f : A \rightarrow C$$

is defined by

$$(g \circ f)(x) = g(f(x))$$

BOARD ALERT

In general,

$$g \circ f \neq f \circ g$$

Composition is generally **not commutative**.

KEY RESULT / FORMULA

Composition is associative:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

11. Inverse Function

A function $f : A \rightarrow B$ has an inverse function

$$f^{-1} : B \rightarrow A$$

if and only if f is bijective.

KEY RESULT / FORMULA

$$f^{-1} \circ f = I_A$$

and

$$f \circ f^{-1} = I_B$$

where I_A and I_B are identity functions.

BOARD ALERT

$$\text{Inverse function exists} \iff f \text{ is bijective}$$

12. How to Find $f^{-1}(x)$

Suppose

$$y = f(x).$$

Then:

1. Write $y = f(x)$.
2. Interchange x and y .
3. Solve for y .
4. The resulting expression is $f^{-1}(x)$.

QUICK EXAMPLE

Let

$$f(x) = 2x + 3.$$

Then

$$y = 2x + 3.$$

Interchange x and y :

$$x = 2y + 3.$$

Therefore,

$$y = \frac{x - 3}{2}.$$

Hence,

$$f^{-1}(x) = \frac{x - 3}{2}.$$

13. Finding Domain Quickly

KEY RESULT / FORMULA

Remember these restrictions:

$$\frac{1}{g(x)} : g(x) \neq 0$$

$$\sqrt{g(x)} : g(x) \geq 0$$

$$\log(g(x)) : g(x) > 0$$

For example,

$$f(x) = \frac{1}{x-2}$$

has domain

$$\mathbb{R} \setminus \{2\}.$$

14. Board Exam Quick Tests

BOARD ALERT

To test whether a relation is a function:

Check the first components.

$$\text{One input} \longrightarrow \text{exactly one output}$$

To test one-one:

Assume

$$f(x_1) = f(x_2)$$

and prove

$$x_1 = x_2.$$

To test onto:

Take arbitrary

$$y \in B$$

and solve

$$y = f(x).$$

If a suitable $x \in A$ exists for every $y \in B$, then f is onto.

To find inverse:

Interchange x and y and solve for y .

15. MUST REMEMBER FOR BOARD

KEY RESULT / FORMULA

$$R \subseteq A \times B$$

$$\# \text{ Relations} = 2^{mn}$$

$$\text{Equivalence Relation} = \text{Reflexive} + \text{Symmetric} + \text{Transitive}$$

$$\text{Function: Every input has exactly one output}$$

$$\text{Onto} \iff \text{Range} = \text{Codomain}$$

$$\text{Bijective} = \text{One-One} + \text{Onto}$$

$$(g \circ f)(x) = g(f(x))$$

$$g \circ f \neq f \circ g \quad \text{in general}$$

$$f^{-1} \text{ exists} \iff f \text{ is bijective}$$

$$f^{-1} \circ f = I, \quad f \circ f^{-1} = I$$

30-SECOND REVISION

$$\text{Relation} \rightarrow \text{Function} \rightarrow \text{One-One/Onto} \rightarrow \text{Composition} \rightarrow \text{Inverse}$$

$$\text{Range} \subseteq \text{Codomain}$$

$$\text{Inverse exists} \iff \text{Bijective}$$

NCERT examples + definitions + composition + inverse = MUST DO for Board revision.

IMPORTANT QUESTIONS

RELATIONS & FUNCTIONS

Board + JEE Main + JEE Advanced

LEVEL 1: BOARD EXAM – MUST DO**B1.** Let

$$A = \{1, 2, 3\}, \quad B = \{a, b\}.$$

Find $A \times B$ and $B \times A$. Hence, verify that

$$A \times B \neq B \times A.$$

B2. If

$$n(A) = 3, \quad n(B) = 4,$$

find the number of elements in $A \times B$ and the number of relations from A to B .**B3.** Let

$$A = \{1, 2, 3\}.$$

Find the number of relations on A .**B4.** Let

$$A = \{1, 2, 3\}$$

and

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}.$$

Check whether R is reflexive, symmetric and transitive.**B5.** Show that the relation R defined on $\mathbb{Z}$ by

$$aRb \iff a - b \text{ is divisible by } 3$$

is an equivalence relation.

B6. Let

$$A = \{1, 2, 3, 4\}$$

and

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)\}.$$

Determine whether R is an equivalence relation.**B7.** Determine whether the relation

$$R = \{(x, y) : x, y \in \mathbb{R}, x \leq y\}$$

is reflexive, symmetric and transitive.

B8. Determine whether

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = 2x + 3$$

is one-one and onto.

B9. Determine whether

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2$$

is one-one and onto.

B10. Find the domain and range of

$$f(x) = \frac{1}{x-2}.$$

B11. Find the domain of

$$f(x) = \sqrt{2x-3}.$$

B12. Find the domain of

$$f(x) = \log(x^2 - 4).$$

LEVEL 2: JEE MAIN**M1.** Let

$$A = \{1, 2, 3, 4\}.$$

The number of reflexive relations on A is

- (a) 2^4
- (b) 2^8
- (c) 2^{12}
- (d) 2^{16}

M2. The number of symmetric relations on a set containing n elements is

$$2^{\left\lfloor \frac{n(n+1)}{2} \right\rfloor}.$$

Find the number of symmetric relations on a set containing 4 elements.

M3. Let

$$f(x) = \frac{x-1}{x+1}.$$

Find

$$f(f(x)).$$

M4. If

$$f(x) = \frac{ax+b}{cx+d}, \quad ad-bc \neq 0,$$

show that f is one-one on its domain.**M5.** If

$$f(x) = x^2 - 4x + 5,$$

find the minimum value of $f(x)$ and hence determine its range.**M6.** Find the range of

$$f(x) = \frac{x^2+1}{x^2+2}.$$

M7. Find the range of

$$f(x) = x + \frac{1}{x}, \quad x \neq 0.$$

M8. Determine the values of a for which

$$f(x) = x^2 + ax + 1$$

is one-one on the interval $[1, \infty)$.**M9.** If

$$f(x) = 2x + 1, \quad g(x) = \frac{x-1}{2},$$

verify that

$$g = f^{-1}.$$

M10. If

$$(f \circ g)(x) = x$$

for every x in the domain of g , does it necessarily imply

$$(g \circ f)(x) = x?$$

JEE ADVANCED CHALLENGE**Question:**

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = x + \sqrt{x^2 + 1}.$$

- (a) Prove that f is one-one.
- (b) Find the range of f .
- (c) Find $f^{-1}(x)$.
- (d) Verify that

$$f^{-1}(f(x)) = x.$$

Hint:

Rationalise:

$$x + \sqrt{x^2 + 1} > 0$$

and use

$$(\sqrt{x^2 + 1} + x)(\sqrt{x^2 + 1} - x) = 1.$$

LAST-MINUTE FORMULA SHEET

$$R \subseteq A \times B$$

$$\#(A \times B) = n(A)n(B)$$

$$\# \text{ Relations} = 2^{n(A)n(B)}$$

$$\text{Reflexive: } (a, a) \in R$$

$$\text{Symmetric: } (a, b) \in R \Rightarrow (b, a) \in R$$

$$\text{Transitive: } (a, b), (b, c) \in R \Rightarrow (a, c) \in R$$

$$\text{Equivalence} = \text{Reflexive} + \text{Symmetric} + \text{Transitive}$$

$$\text{One-One: } f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

$$\text{Onto: Range} = \text{Codomain}$$

$$\text{Bijective} = \text{One-One} + \text{Onto}$$

$$(g \circ f)(x) = g(f(x))$$

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

$$f^{-1} \text{ exists} \iff f \text{ is bijective}$$